

RETRIEVAL-AUGMENTED GENERATION CHATBOT

Healthcare Insurance Assistant

Grounded, cited answers about a health plan — built strictly from the plan's own documents, never from the model's general training knowledge.

Grounded & Cited

Hybrid Retrieval

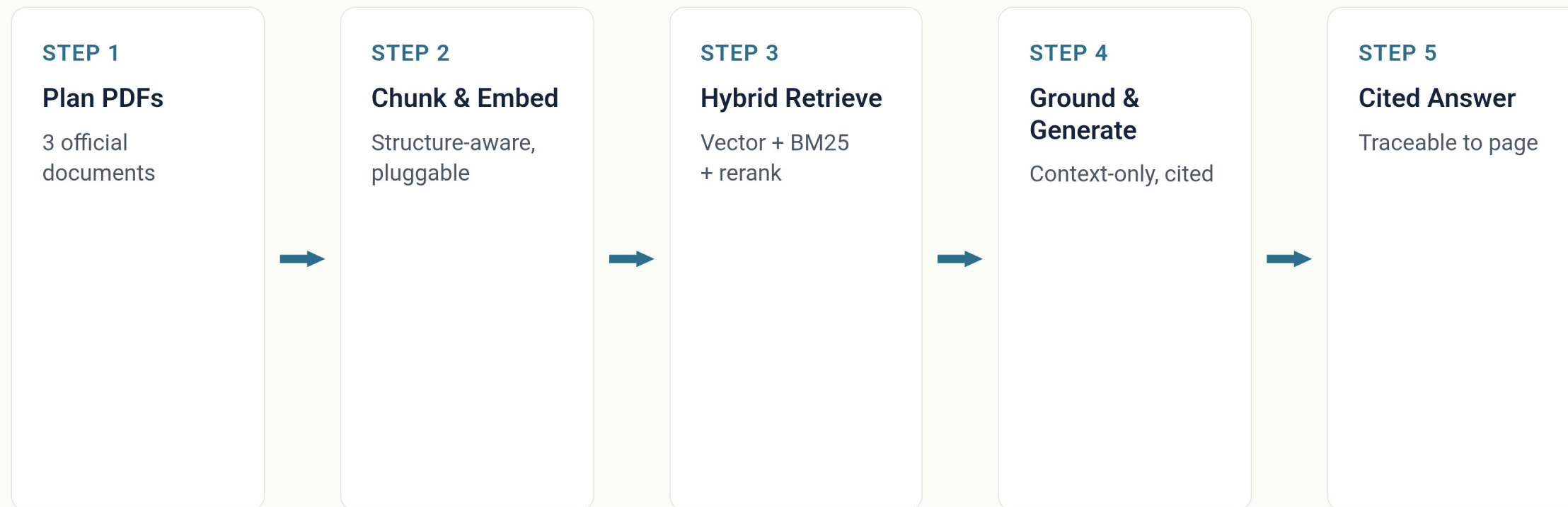
Guardrails

Keyless Production Path

THE PIPELINE

From Plan PDFs to Grounded, Cited Answers

An insurance question needs an answer traceable to the plan's own documents, not a fluent guess from the model's general training data. Every stage below exists to enforce that.



Implemented in rag_pipeline/ and retrieval_service.py

WHY THIS ISN'T A TOY DEMO

What Makes This Production-Grade



Grounded & Cited

Every claim cites a document and page;
exact fallback otherwise.



Hybrid Retrieval + Reranking

Vector + BM25 blend, cross-encoder
rerank — measured, not assumed.



Defense-in-Depth Guardrails

Input denylist, strict grounding prompt,
output lexical check.



Measured Retrieval Quality

MRR 0.950 on the real corpus;
Recall/NDCG before vs. after rerank.



Full Observability

Every turn traced in MLflow; thumbs-
up/down feedback loop.



Pluggable, Keyless-Ready

Local, Gemini, or Databricks — one
interface, zero rewrites.

HONEST ABOUT THE GAPS

Known Gaps & the Path to Production

Known Gaps

- ⚠ No PHI/PII redaction layer — flagged, not solved
- ⚠ Citations aren't verified against retrieved chunks
- ⚠ Capped 6-turn memory loses early conversation context

Path to Production

- 🔑 Keyless Databricks Foundation Model APIs — implemented
- 📄 ChromaDB + SQLite to Vector Search + Delta, same interfaces
- 📊 MRR/Recall@K harness makes tuning measurable, not guesswork

Thank you — happy to go deeper on any layer.